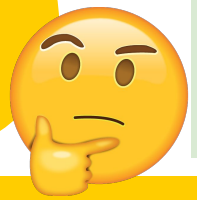
Basic circuits

Lesson objectives

- Draw and interpret circuit diagrams, including all common circuit symbols
- Identify faulty circuits and identify how to fix them

Brain teaser



1. What is an atom?
2. Three sub-atomic particles are found within an atom, can you name these particles?
3. Do you know the charges that these three sub-atomic particles possess?

Key words: current, switch, ammeter, cell, battery, motor

Success criteria

Electric circuits		
Working towards	Working at	Working beyond
I can recognise various circuit symbols.	I can draw circuit diagrams using the correct symbols.	I can describe current as the rate of flow of charge.



Success criteria

Lesson Title	Meeting	Exceeding	Mastering
Electric Circuits	☐ I can recognise various circuit symbols	☐ I can draw circuit diagrams using the correct symbols	☐ I can describe current as the rate of flow of charge
Series and Parallel Circuits	☐ I can recognise series and parallel circuits	☐ I can explain why it would be useful to use a parallel circuit in some situations	☐ I can explain why voltmeters are added parallel and not in series.
Current in Series and Parallel Circuits	☐ I can describe the role of an ammeter	☐ I can explain how current differs in series and parallel circuits	☐ I am able to predict the current in series and parallel circuits
Voltage in Series and Parallel Circuits	☐ I can describe the role of a voltmeter	☐ I can explain the effect series and parallel circuits have on potential difference	☐ I can explain why voltmeters are added parallel to a circuit, not in series
Resistance	☐ I can definite the term resistance	☐ I can explain what causes resistance in a circuit	☐ I can use the equation $V=IR$ and rearrange it confidently to calculate voltage, current, or resistance in a circuit.

Electricity

Electricity is not one single thing - It is a word we use for a particular transfer of energy.

So what does this mean?

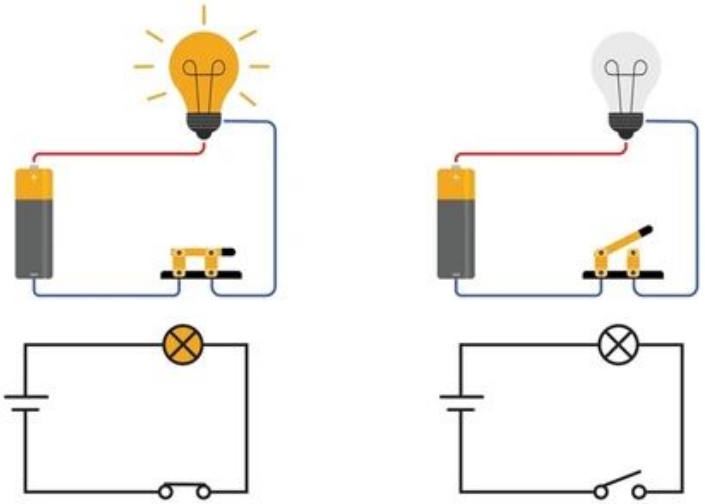
What we know as electricity is actually **a flow of electrons around a circuit**.

The only time you can use the word ELECTRICITY is in the title of this unit!

Circuit symbols

Electrical circuits are often represented by circuit diagrams.

Rather than spending lots of time creating very artistic drawings of circuits, we use standard symbols that are much **simpler, easier to draw and interpret**.



Circuit symbols

What are the circuit symbols for the following components?

- wire
- bulb 
- switch 

Battery or cell?



Cell (Single unit)

Battery (Multiple units)





A cell can **store chemical energy** which **can be transferred into electrical energy**

A battery is **many cells put together** in series

Circuit symbols

Component	Symbol	Function
lamp		Lights up when current flows through it

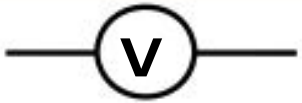


Complete the worksheet





Circuit symbols - you need to make sure you know the component, symbol and function!

Component	Symbol	Function
lamp		Lights up when current flows through it
fuse		Breaks a circuit if too much current flows through it by the metal wire melting
open switch		Creates a break in the circuit so that current cannot flow
diode		Allows current to flow through it in one direction only
voltmeter		Measures potential difference across a component in volts

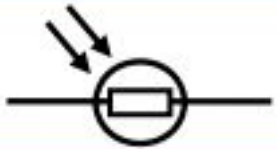
Fill up the worksheet given or make sure it's written in your book

Circuit symbols - you need to make sure you know the component, symbol and function!

cell		Source of potential difference in a circuit
resistor		Has a designed resistance to electric current passing through it
thermistor		A resistor which reduces in resistance when heated
LED		Diode that emits light when a voltage is applied
closed switch		Completes a circuit so that electric current can flow
battery		Two or more cells electrically connected to provide a source of potential difference

Fill up the worksheet given or make sure it's written in your book

Circuit symbols - you need to make sure you know the component, symbol and function!

LDR		A resistor whose resistance varies depending on the light intensity
ammeter		Measures electric current through a circuit in amperes
variable resistor		A component which can have its resistance changed to vary the amount of current flowing through a circuit

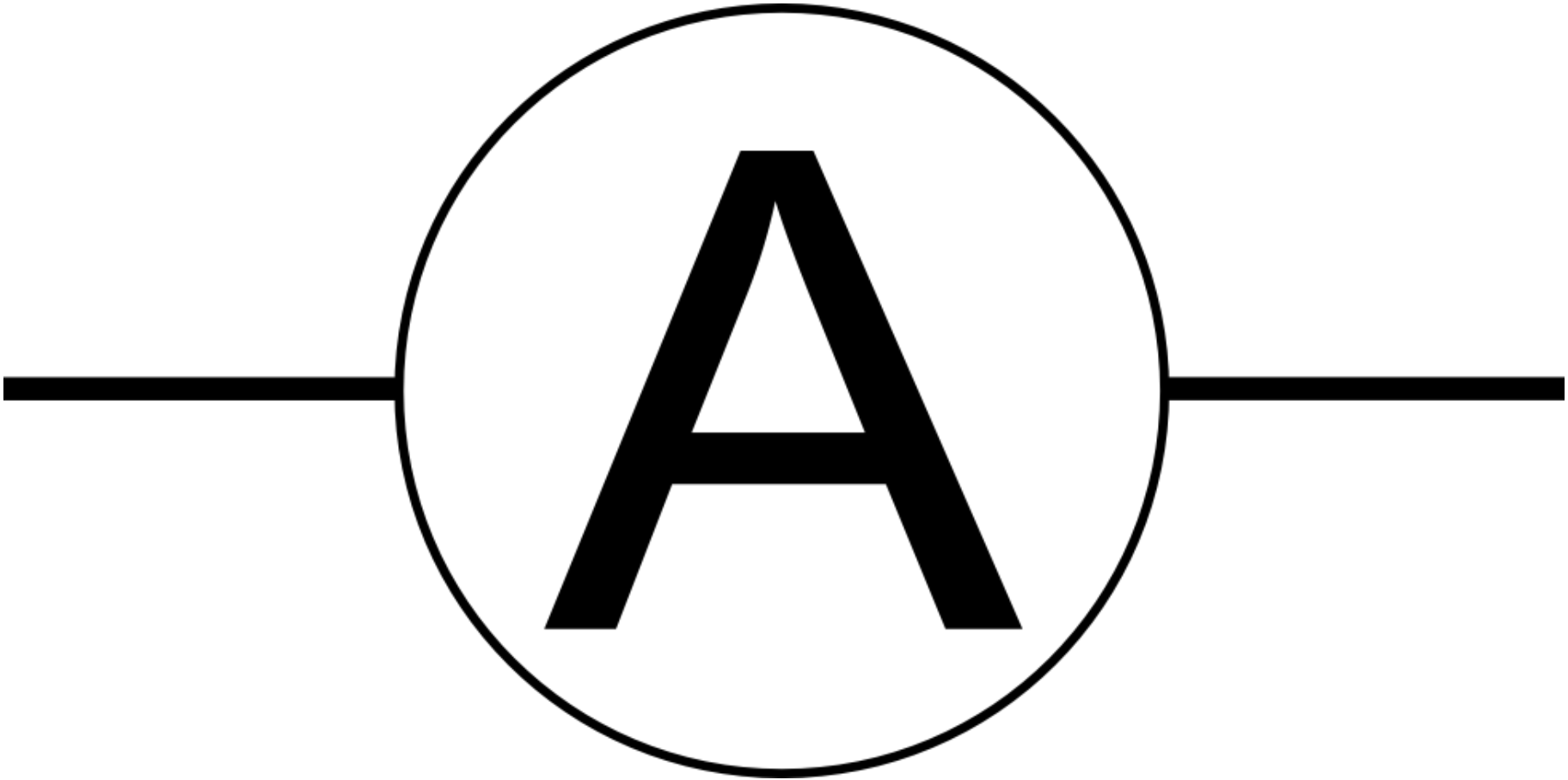
Fill up the worksheet given or make sure it's written in your book


What
am I?

What am I?



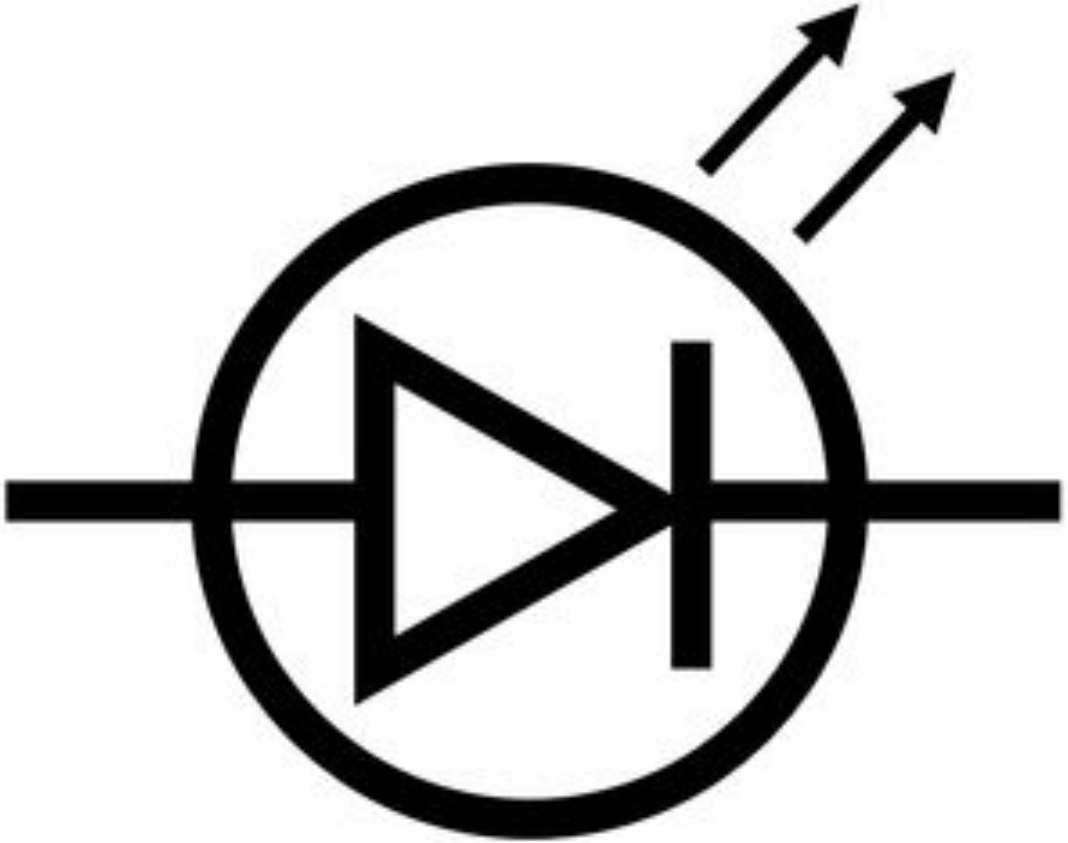
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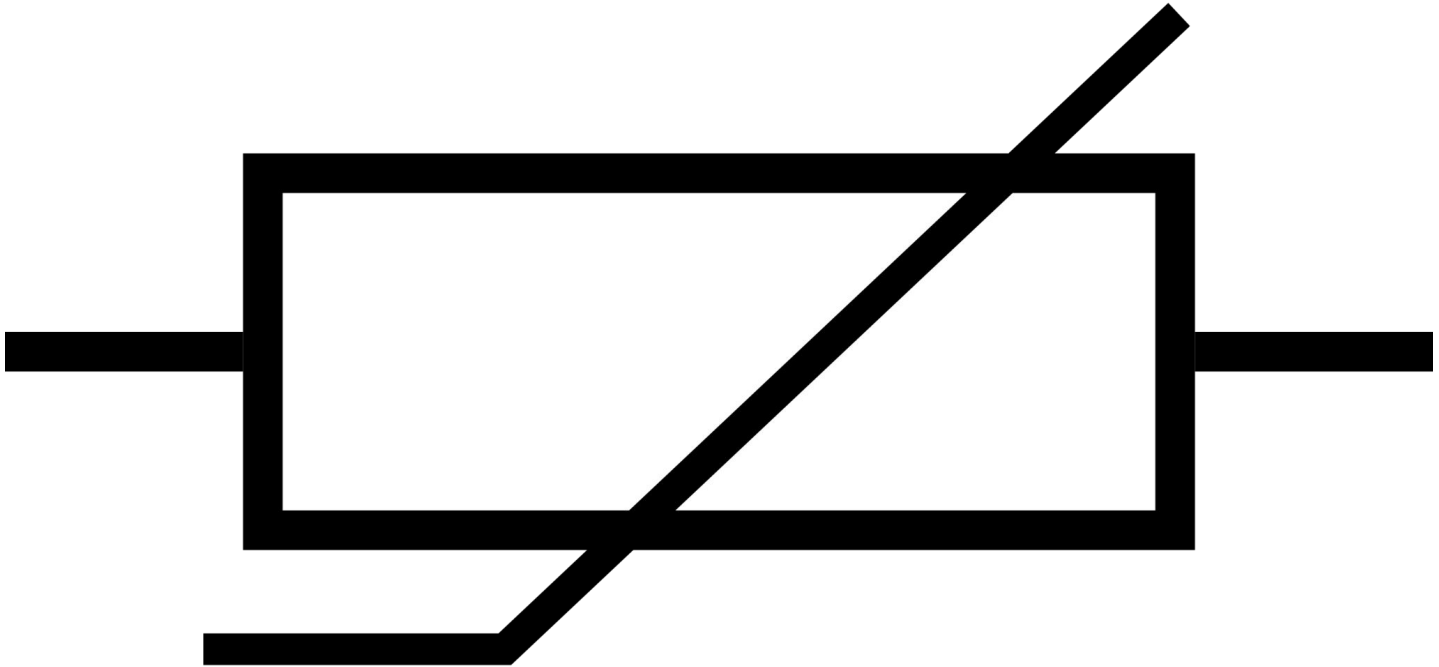
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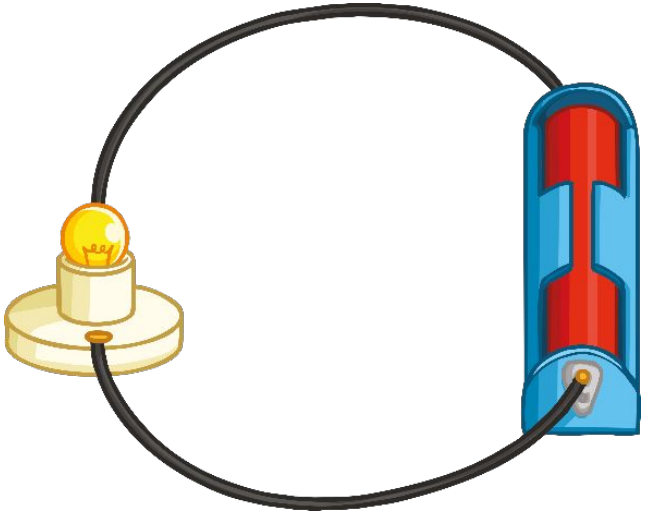
What am I?



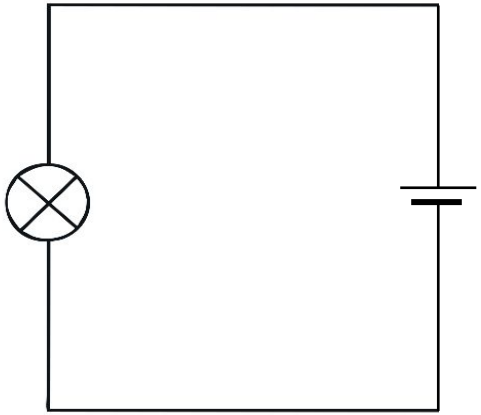
What am I?



Circuit symbols



How would you begin to draw this circuit?





List 5 components to include in a circuit

Swap your book with a partner who will draw the circuit

Use a pencil and ruler to draw circuits with straight lines



A student connects the electrical circuit shown in Fig. 9.1.

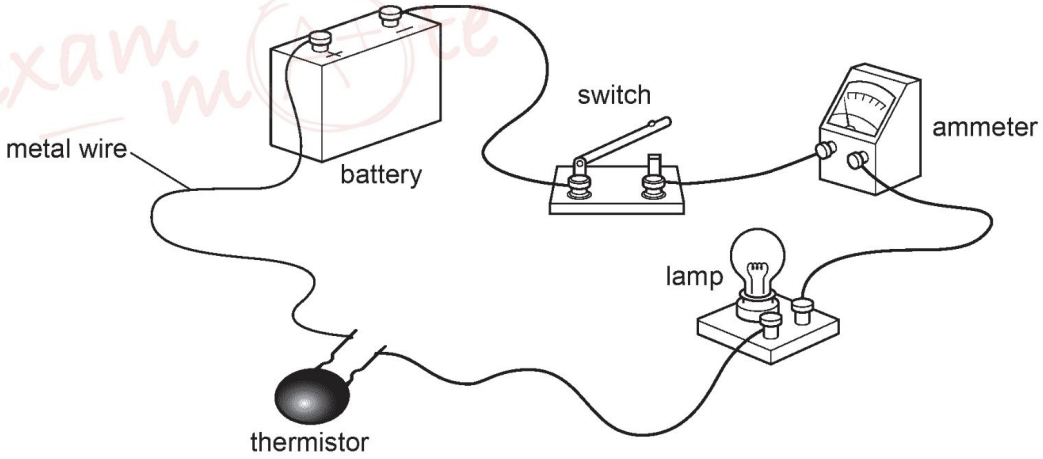


Fig. 9.1

Fig. 9.2 shows part of the circuit diagram for the circuit in Fig. 9.1.

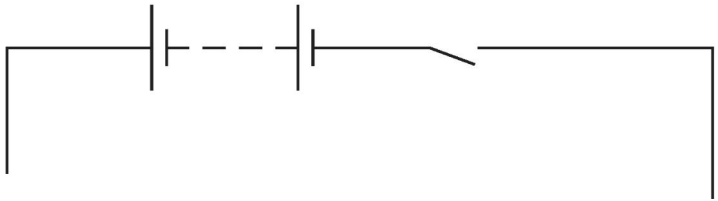


Fig. 9.2

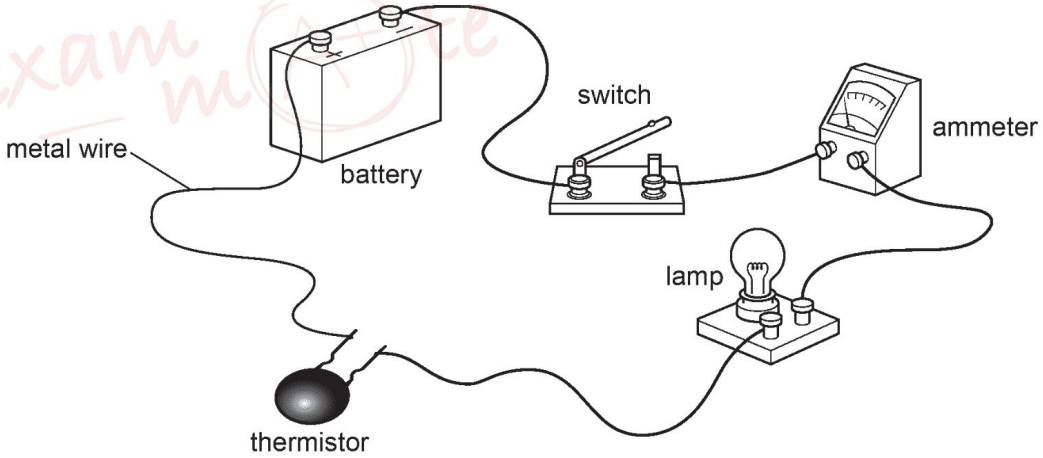
Complete the circuit diagram in Fig. 9.2 to represent the circuit in Fig. 9.1. Use standard electrical symbols.

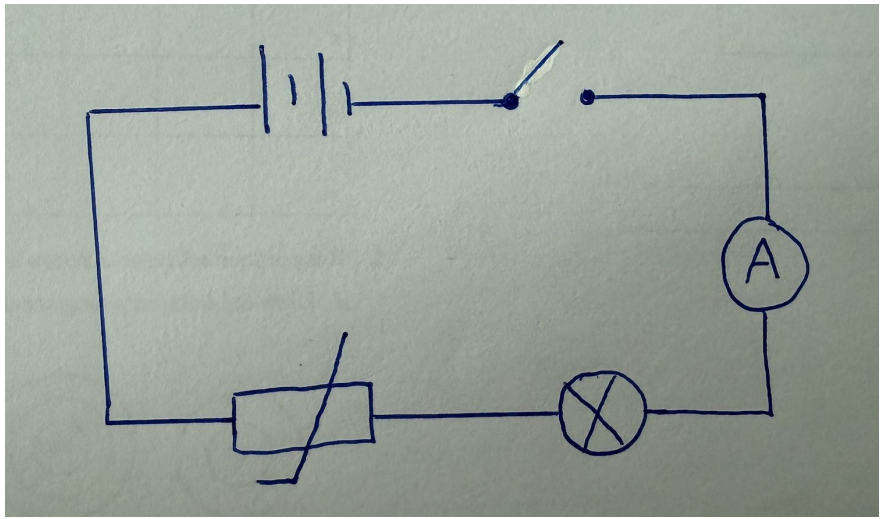
[4]

Use a pencil and ruler to draw circuits with straight lines



A student connects the electrical circuit shown in Fig. 9.1.

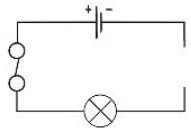
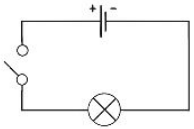
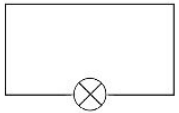
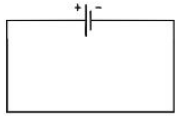




Use a pencil and ruler to draw circuits with straight lines

Find the fault

Find the Fault

Fault Finding Circuit Number	Will the bulb light? yes/no	Where is the fault on the circuit? Circle the fault.	Draw the correct circuit that will light the bulb.	Describe how you fixed the fault to light the bulb.
1				
2				
3				
4				

Science



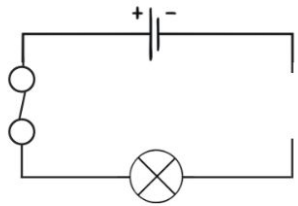
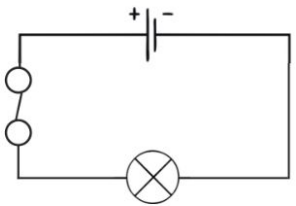
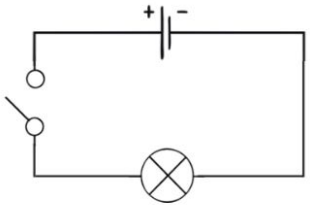
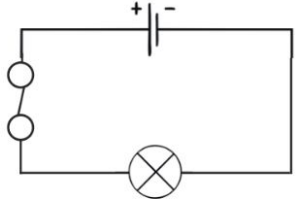
Identify the fault and say whether you think the bulb will light or not.

Use a pencil and ruler to draw circuits with straight lines

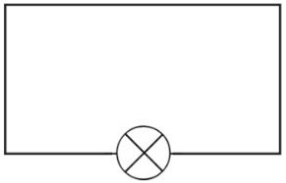
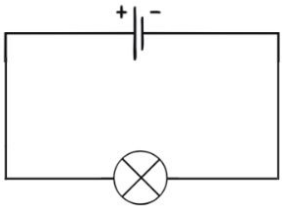
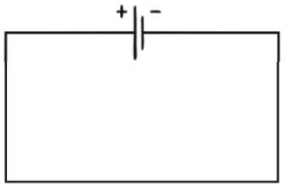
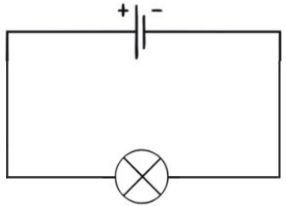


Mark your work using a green pen!

Find the Fault Answers

Fault Finding Circuit Number	Will the bulb light? yes/no	Where is the fault on the circuit? Circle the fault.	Draw the correct circuit that will light the bulb.	Describe how you fixed the fault to light the bulb.
1	no			Student's answers may vary. I placed a wire between the cell and bulb to complete the circuit.
2	no			Student's answers may vary. I closed the switch to complete the circuit.

Mark your work using a green pen!

3	no			Student's answers may vary. I placed a cell in the circuit.
4	no			Student's answers may vary. I placed a bulb in the circuit.

Success criteria

Lesson Title	Meeting	Exceeding	Mastering
Electric Circuits	☐ I can recognise various circuit symbols	☐ I can draw circuit diagrams using the correct symbols	☐ I can describe current as the rate of flow of charge
Series and Parallel Circuits	☐ I can recognise series and parallel circuits	☐ I can explain why it would be useful to use a parallel circuit in some situations	☐ I can explain why voltmeters are added parallel and not in series.
Current in Series and Parallel Circuits	☐ I can describe the role of an ammeter	☐ I can explain how current differs in series and parallel circuits	☐ I am able to predict the current in series and parallel circuits
Voltage in Series and Parallel Circuits	☐ I can describe the role of a voltmeter	☐ I can explain the effect series and parallel circuits have on potential difference	☐ I can explain why voltmeters are added parallel to a circuit, not in series
Resistance	☐ I can definite the term resistance	☐ I can explain what causes resistance in a circuit	☐ I can use the equation $V=IR$ and rearrange it confidently to calculate voltage, current, or resistance in a circuit.